Pablo Ibáñez-Porras

Predoctoral Researcher at CISA-INIA-CSIC

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PROFESSIONAL PROFILE

Predoctoral researcher specialising in spatial epidemiology, mathematical modelling, data science and geographic information systems for One Health surveillance and risk assessment.

RESEARCH PROJECT EXPERIENCE

2024 - 2026 **Development of models, viewers and applications for rapid risk assessment of emerging pathogens**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded by the European Partnership on Animal Health and Welfare Horizon Europe GA 101136346 (2024-2027)

2023 - 2024 **Cost-benefit model of early wild-boar vaccination following African swine fever (ASF) outbreaks**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded by the VACDIVA project Horizon 2020 GA 862874 (2023-2024)

2022 - 2023 **Enhancement of the DiFLUsion warning system for avian-influenza introduction by wild birds in Spain and development of ProtectIA to assess introduction risk in poultry holdings**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded through a management agreement with the Spanish Ministry of Agriculture, Fisheries and Food AEG21-198 (2022-2023)

EDUCATION

2025 - present **PhD in Computer Engineering (ongoing)**
University of Salamanca
Programme jointly supervised by Irene Iglesias (Animal Health Research Centre, CISA-INIA-CSIC) and Ángel Martín del Rey (University Institute of Fundamental Physics and Mathematics, IUFFYM-USAL).
Development and application of an integrated epidemiological risk-assessment model using mathematical and computational modelling to analyse the factors associated with the introduction and spread of HPAI in wild birds and estimate transmission risk to livestock holdings.

2021 **Master's Degree in Mathematical Modelling**
University of Salamanca
MSc thesis: 'Epidemic-spread model between communities defined on complex networks considering population flow'. View project

2021 **Monte Carlo modelling of proton-beam interactions with different targets**
Centre for Pulsed, Ultrashort and Ultraintense Lasers (CLPU)

2020 **BSc in Physics**
University of Salamanca
BSc thesis: 'Complex networks: description, analysis and applications'. View project

SCIENTIFIC PUBLICATIONS

Abdrakhmanov, S.K., Iglesias, I., Ibañez-Porras, P., Mukhanbetkaliyev, Y.Y., Mukhanbetkaliyeva, A.A., Ruzmatov, S.I., Kadyrov, A.S., Balgabekova, K.T. (2026). Development of a dashboard for monitoring and risk assessment of highly pathogenic avian influenza in Kazakhstan. Central Asian Journal of Veterinary Science. https://doi.org/10.51452/cajvs.2026.2(014).2224

Duarte-Benvenuto A, Sacristán C, Ewbank AC, Costa-Silva S, Groch KR, Díaz-Delgado J, Ibáñez-Porras P, Colosio AC, da Cunha Gomes Ramos H, Carvalho VL, Ribeiro VL, Bertozzi CP, Pessi CF, Gravena W, da Silva VMF, Marmontel M, Marigo J, Catão-Dias JL. (2026). Molecular surveillance of flaviviruses in wild cetaceans from Brazil.. Journal of Wildlife Diseases. https://doi.org/10.7589/JWD-D-25-00149

Ibañez-Porras, P., Bosch, J., Feliziani, F., Maresca, C., Sanchez-Vizcaino, J. M., Iglesias, I., et al., Martinez Aviles, M. (2025). WiBISS: A tool to estimate avoided lost revenue of African swine fever wild boar vaccination at municipality level. Frontiers in Veterinary Science. https://doi.org/10.3389/fvets.2025.1667173

Iglesias, I., Ruzmatov, S., Ibañez-Porras, P., Kadyrov, A., Bakishev, T., Korennoy, F., Perez, A., Abdrakhmanov, S. (2025). Risk Assessment of Highly Pathogenic Avian Influenza (HPAI) in Wild Bird Species in Kazakhstan.. German Journal of Veterinary Research. https://doi.org/10.51585/gjvr.2025.3.0149

Ibáñez-Porras, P., de la Torre, A., Gómez-Pérez, J. I., Tomás-Tenllado, C., García, E., Cáceres, G., Pérez, A., Iglesias, I. (2024). DiFLUsion: A novel space-time alert system for HPAI.. German Journal of Veterinary Research. https://doi.org/10.51585/gjvr.2024.4.0107

Sacristán C, Ewbank AC, Ibáñez Porras P, Pérez Ramírez E, de la Torre A, Briones V, Iglesias I. (2024). Novel epidemiologic features of highly pathogenic avian influenza virus A H5N1 2.3.3.4b panzootic: a review.. Transboundary and Emerging Diseases. https://doi.org/10.1155/2024/5322378

Ewbank AC, Catão-Dias JL, Navas-Suarez PE, Duarte-Benvenuto A, Zamana-Ramblas R, Ferreira-Machado E, Christino Lial H, Ibañez-Porras P, Sacristán I, Sacristán C. (2024). Novel alpha-, beta-, and gammaherpesviruses in Neotropical carnivores of Brazil.. Transboundary and Emerging Diseases. https://doi.org/10.1155/2024/1347516

Duarte-Benvenuto A, Sánchez-Sarmiento AM, Ewbank AC, Zamana-Ramblas R, Costa-Silva S, Silvestre N, Faita T, Keid LB, Soares RM, Pessi CF, Sabbadini JR, Borges MF, Ferioli RB, Marcon M, Barbosa CB, Fernandes NCCA, Ibáñez-Porras P, Navas-Suárez PE, Catão-Dias JL, Sacristán C. (2024). Bacterial septicemia and herpesvirus infection in Antarctic fur seals (Arctocephalus gazella) stranded in the São Paulo coast, Brazil.. Veterinary Research Communications. https://doi.org/10.1007/s11259-024-10408-x

Duarte-Benvenuto A, Sacristán C, Ewbank AC, Zamana-Ramblas R, Lial HC, Silva SC, Arias Lugo MA, Keid LB, Pessi CF, Sabbadini JR, Ribeiro VL, do Valle RDR, Bertozzi CP, Colosio AC, Ramos HDCG, Sánchez-Sarmiento AM, Ferioli RB, Pavanelli L, Ikeda JMP, Carvalho VL, Catardo Gonçalves FA, Ibáñez-Porras P, Sacristán I, Catão-Dias JL. (2023). Molecular Detection and Characterization of Mycoplasma spp. in Marine Mammals from Brazil.. Emerging Infectious Diseases. https://doi.org/10.3201/eid2912.23090

TECHNICAL AND KNOWLEDGE-TRANSFER ARTICLES

Iglesias Irene; Lim Seunghyun,. de la Torre Ana, Kanankege Kaushi, Ibañez Pablo, Fillman Kathryn, Blanco Carlos E, Perez Andres M (2023). DashFLUboard: a web-based tool for assessing and predicting global spread of highly pathogenic avian influenza in near real time. EsriNews. Link

Irene Iglesias, Elena García, Germán Cáceres, Pablo Ibañez, José Ignacio Gómez, Christian Tomas-Tenllado,Carlos Eduardo Blanco Urbina, Ana de la Torre (2023). Influenza Aviar: cambios en la dinámica de la enfermedad y cómo anticiparnos a ella mediante sistemas de vigilancia a tiempo real. Avinews. Link

SCIENTIFIC ACTIVITY

26 scientific events recorded · 20 lead-author or speaker contributions · 14 international contributions.

Lead-author contributions

- SMART-E: A Spatial Digital Framework for Standardized Environmental AMR Surveillance - Oral; European One Health Association Annual Scientific Meeting 2026 (EOHA 2026) (2026)
- From Science to Policy: Co-developing a Real-Time One Health Early Warning System for Avian Influenza (DiFLUsion) - Poster; European One Health Association Annual Scientific Meeting 2026 (EOHA 2026) (2026)
- ASF FrontWave: A Rapid Risk Assessment interactive mapping tool - Poster; SVEPM Conference 2026+ (2026)
- Mapping anthropogenic sources of antimicrobial-resistance emissions - Oral; Esri Spain Conference 2026 (2025)
- ProtectIA: A geospatial tool for avian-influenza prevention - Oral; 4th International Young Researchers Meeting INNOVA-Salamanca (2025)
- Geoprocessing tools to support emerging-disease response - Oral; 1st Interdisciplinary Congress on Medicine and Engineering (2025)
- WiBISS: A tool to estimate economic benefits of African swine fever wild boar vaccination for pig producers - Poster; GARA Scientific Meeting (2025)
- DiFLUsion: a tool for surveillance against Avian Influenza - Oral; 2nd General Assembly of the CSIC Computational Biology and Bioinformatics Network (Conexión BCB) (2025)
- Epidemiology and spatial analysis: using GIS to understand disease dynamics and manage public and animal health from a One Health perspective - Oral; Esri Spain Conference 2024 (2024)
- Soil contamination by antibiotics and antimicrobial-resistance genes - Poster; 10th CONDEGRES Symposium (2024)
- MarineStrandingViewer: An Online Tool to Record Marine Mammal Strandings - Poster; IAAAM 55th Annual Conference (2024)
- Real-time avian-influenza warning system: DiFLUsion - Poster; Health Surveillance Meeting 2023: present and future towards 2030 (2023)
- Ants, chickens and mathematics - Oral; INIAciando ciencia (2023)
- DiFLUsion avian-influenza surveillance system: direct use of data from SEO/BirdLife citizen-science programmes - Poster; Spanish Ornithology Congress (2022)

TEACHING, ADVICE AND KNOWLEDGE TRANSFER

- The overlooked role of Passeriformes in avian influenza and antimicrobial resistance amid climate change - Workshop LINCGLOBAL Scientific and Communication Training (2025)
- Training course on Risk analysis for transboundary animal disease control - World Organization for Animal Health (WOAH) Sub-Regional Representation for South-East Asia (2025)
- Implementing AI in animal-health research - Workshop on Artificial Intelligence, Big Data and digitalisation in veterinary medicines (2025)
- Mentor for the Scientists in Practice programme - AEAC / CSIC - CISA (2024)
- Analysis of the spatio-temporal distribution of several animal diseases - External-placement supervision (2023)
- Mathematical tools for temporal smoothing of migration routes - Bachelor's thesis supervision (2023)
- Analysis of the spatio-temporal distribution of HPAI outbreaks in Spain - Universidad Complutense de Madrid (2022)
- Technical advice on the highly pathogenic avian influenza (HPAI) situation and the use of DiFLUsion as an early-warning system (2022-2023).

SELECTED CONTINUING EDUCATION

Selected courses from 607 accredited hours of continuing education.

- Biodiversity analysis with QGIS, Google Earth Engine and OpenData - 125 h · GeoInnnova
- Advanced Python - 40 h · CSIC
- Geoprocessing scripts with Python in ArcGIS Pro - 40 h · Esri España

- Machine learning for scientific research: random forests, boosting and advanced interpretation - 24 h · Museo Nacional de Ciencias Naturales (MNCN-CSIC)
- Deep learning with Keras for scientific research - 21 h · Museo Nacional de Ciencias Naturales (MNCN-CSIC)
- Web development with the ArcGIS API for JavaScript - 25 h · Esri España
- Services with Docker - 20 h · CSIC (Plan de Formación)
- Adapted numerical methods for differential equations - 10 h · University of Salamanca
- FME and databases - 8 h · Consejo Superior de Investigaciones Científicas (CSIC)

COLLABORATION IN TOOL DEVELOPMENT

- DiFLUsion - Spatio-temporal warning system for highly pathogenic avian influenza
- ProtectIA - Multi-criteria geospatial tool for avian-influenza prevention
- SMART-E - Surveillance and Monitoring of Antimicrobial Resistance in the Environment
- WiBISS - Wild Boar Immunization Strategy Simulator
- ASF FrontWave - African Swine Fever Front-Wave Rapid Risk Assessment
- Dashboards - Epidemiological dashboards for surveillance and risk assessment

Across these tools, my contribution has included the development and mathematical formalisation of the models, the integration of spatial and temporal data, the implementation of computational analysis algorithms, and the interpretation of results.

SCIENCE OUTREACH

- Epidemic detectives: chickens and mathematics - Festival Pint of Science (2025)
- Mentor for the Scientists in Practice programme - (2024)
- Epidemics and pandemics: become the most dangerous virus - Semana de la Ciencia y la Tecnología (2022)
- Become a real epidemic detective - Semana de la Ciencia y la Tecnología (2022)
- Epidemics and pandemics: become the most dangerous virus - Noche Europea de los Investigadores (2022)

AWARDS, FELLOWSHIPS AND RESEARCH STAYS

- MOVTEC 2025-12 research stay · Consejo Superior de Investigaciones Científicas (CSIC) · 2026
- 2024 Research Award · Instituto de Estudios del Huevo · 2024
- Introduction to Research Fellowship · Instituto Universitario de Física Fundamental y Matemáticas (IUFFYM) · 2021

SKILLS

- Python, MATLAB, Mathematica and Fortran
- QGIS, ArcGIS Pro/Online, Google Earth Engine and remote sensing
- Mathematical modelling, spatio-temporal analysis and machine learning
- Docker, geospatial application development and data visualisation
- LaTeX and Microsoft Office
- Professional English